Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Affordable Housing & Leasing Compliance · Agentic AI · LangGraph Multi-Agent · HITL & Confidence Scoring · LLM Evaluation · Python

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PROFESSIONAL SUMMARY

8 years building affordable housing compliance systems at Entrata — LIHTC, Section 8/HCV, HUD, HAP, AMI — for exactly the low-to-moderate income renter population Boom serves. Built **Leasel**: a leasing compliance AI platform with an 8-agent LangGraph pipeline, confidence-gated HITL escalation, and a core rule enforced in code: *LLM reasons, deterministic logic or a human makes every regulated eligibility and compliance call.* Shipping AI with Claude Code daily. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Affordable Housing Compliance (LIHTC, HUD, HAP, AMI) | Leasing & Underwriting Agent Design | Agentic AI Systems | LangGraph Multi-Agent | HITL & Confidence-Gated Escalation | Deterministic Guardrails in Code | LLM Eval Suite Ownership | RAG Pipelines & Retrieval | Claude Code Daily Use | Anthropic & OpenAI APIs | Production Python | CI-Integrated Agent Evals

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

ML Research · Statistical Modeling · Python Instruction

- Built prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection surfacing data integrity issues manual review missed; mentored 600+ students across Python programming and database design.

Entrata (PropTech / Residential Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Affordable Housing Compliance · LIHTC · HUD · HAP · AMI · Production AI

- **8 years serving low-to-moderate income renters** across affordable housing programs: automated income eligibility, AMI threshold calculations, and recertification workflows for **Section 8/HCV, LIHTC, HUD, and HAP compliance** — zero compliance violations across 100K+ unit portfolios; the exact renter population Boom is building for.
- Managed end-to-end financial services for affordable housing: **HAP allocation, rent pricing strategy, AMI-based pricing tiers, tax credit calculations, HUD payments, rental subsidies, and income limits** — analytical risk analysis ensuring compliance with federal and state housing regulations.
- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated complex data workflows with output validation, quality gates, and compliance controls enforced from day one.
- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, RabbitMQ) for 100K+ monthly active users across 120+ Agile sprints — 99.5% uptime; led microservices architecture redesign (Docker), 100% PHI compliance on AWS S3 migration (Libsodium).
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; optimized 200+ SQL queries (30% gain); mentored 5+ engineers across 300+ peer code reviews.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Backend · Affordable Housing Data · AWS S3

- Built affordable housing and property management modules (PHP, PostgreSQL, JavaScript); processed 1.6M HUD/Tax Credit income limit records; created 56 state-specific housing certification documents; migrated client documents to AWS S3.

PORTFOLIO PROJECTS

Leasel — Leasing Compliance AI Platform: 8-Agent LangGraph + Confidence-Gated HITL

Leasing Agent | HITL | Guardrails

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built **for exactly the leasing and compliance workflow Boom is automating**: an 8-agent LangGraph StateGraph (Triage → Retrieval → Config → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) for LIHTC/affordable housing lease compliance — real conditional routing, confidence scoring at every stage, HITL escalation when threshold not met.
- **Core design rule enforced in code**: LLM extracts and reasons; deterministic logic or a human makes every regulated eligibility and compliance call — never reversed. 0 safety boundary violations on full validation set; 100% citation correctness.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants); pass@1 81.7% and pass@3 88.3% via pass@k harness with multi-trial sampling, regression gate, and LLM-as-judge scoring.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Orchestration, HITL, Human-in-the-Loop, Confidence Scoring, Guardrails, Deterministic Compliance Gates, Tool Calling, Agent Workflow Integration, Context Engineering, Ethical AI, Production AI

LLM Evaluation & RAG: LLM Evaluation, LLM-as-Judge, Eval Frameworks, CI-Integrated Evals, RAG Pipelines, Semantic Search, Embeddings, ChromaDB, Vector Database, Multi-Provider LLM Routing, Anthropic API, OpenAI API, Groq, Ollama, Prompt Engineering, GenAI

Backend & Infrastructure: Python (primary), PHP (OOP server-side — Ruby/PHP architecturally similar), FastAPI, REST APIs, React, TypeScript, JavaScript, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, Microservices, ETL, RabbitMQ, Linux, Git

Domain & Tooling: Affordable Housing Compliance (LIHTC, Section 8/HCV, HUD, HAP, AMI), Leasing & Underwriting Workflows, Income Eligibility, Certification Compliance, Claude Code (daily use), Agile/Scrum, PyTorch, Scikit-learn, NLP, spaCy, BERT, Statistical Modeling